

Pandu Ranga Reddy Konala

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Personal Statement

I am a cybersecurity researcher specializing in AI-generated code security and Infrastructure as Code quality. My research develops comprehensive framework that detect vulnerabilities ranging from simple misconfigurations to complex supply chain attacks, perform metadata-driven analysis revealing hidden diffusion pathways, and evaluate AI-generated code using zero-shot methods—achieving 100% CIS compliance while improving foundation models code quality by 45.7%. Through systematic analysis of thousands of Infrastructure as Code scripts, I've established scalable detection and remediation methodologies. I seek to advance cybersecurity research, developing innovative approaches that enhance code generation safety and establish standards for AI-assisted development.

Academic Progress

The University of Waikato

PhD in Computer Science

- Fully funded by MBIE, New Zealand, through the AI for Human-Centric Security Catalyst project scholarship
- Research Topic: A Code Quality Framework for Infrastructure as Code in the Modern DevSecOps Era

Hamilton, New Zealand

April 2022 - January 2026

Lancaster University

Masters in Cyber Security

- Fully funded by Lancaster University through the Cyber Security Scholarship program
- Certified by the National Cyber Security Centre (NCSC), UK

Lancaster, United Kingdom

September 2019 - September 2020

Amrita Vishwa Vidyapeetham

Bachelor of Technology in Computer Science and Engineering

- Specialized in Cyber Security

Coimbatore, India

June 2015 - April 2019

Work Experience

PhD Researcher

School of Computing & Mathematical Sciences, The University of Waikato

- Conducting experiments and contributing to cybersecurity-related research activities.
- Providing comprehensive support to faculty and students in research projects and academic activities.

Hamilton, New Zealand

February 2021 - January 2026

Software Developer

Quantum Base

- Designed and implemented secure system architectures to integrate quantum PUFs with IoT devices, bridging the gap between quantum technology and everyday applications.

Lancaster, UK

June 2020 - September 2020

Security Analyst Intern

Infosec Future Pvt Ltd

- Conducted website penetration testing, vulnerability scanning, and risk analysis using automated tools and manual techniques during an eight-week internship.
- Performed detailed information gathering to identify security risks and vulnerabilities.

Lucknow, India

May 2018 - June 2018

Red Hat System Administrator Intern

Complete Open Source Solutions (COSS)

- Gained hands-on experience in managing and maintaining Red Hat Enterprise Linux systems during a four-week internship.
- Handled system installation, configuration, performance monitoring, security management and maintained system documentation.

Hyderabad, India

May 2017

Ethical Hacker Intern

Star Computers

- Gained hands-on experience in penetration testing and vulnerability assessments over a four-week period.
- Identified potential vulnerabilities and security risks in client systems and applications through various tests and assessments.

Vizag, India

May 2016

Professional Certifications

Red Hat Certified Engineer (RHCE)

by Red Hat, License No. 170-138-911 (2017 - 2020)

Red Hat Certified System Administrator (RHCSA)

by Red Hat, License No. 170-138-911 (2017 - 2020)

Certified Ethical Hacker (CEH)

by EC-Council, License No. ECC022209621222 (2016 - 2019)

Skills

Cyber Security Tools

Metasploit Framework, Nessus, Burp Suite & Other Cyber Security Tools

Programming

Python, Java, R, HTML/CSS, SQL, Node.js

Operating Systems

Red Hat Linux, Kali, Debian

Virtualization Technologies

VMware vSphere, Containers (Docker)

Miscellaneous

TeX(Overleaf/ Markdown), Tableau, Microsoft Office, Power Automate, Git, Hack The Box

Peer-Reviewed Research Publications

A Policy-Driven Framework for Measuring the Code Quality of Ansible Roles: An Empirical Study

Under Review

Research Paper (First-Author)

2026

- Journal of Systems and Software (JSS)

Evaluating and Preventing Security Smells in AI-Generated Ansible Code

Under Review

Research Paper (First-Author)

2026

- IFIP Information Security & Privacy Conference (SEC), Australia

Tracking Security Smell Diffusion Patterns in Ansible Playbooks Using Metadata

Published

Research Paper (First-Author)

November 2025

- International Workshop on Security (IWSEC) 2025, Japan

Metadata Assisted Supply-Chain Attack Detection for Ansible

Published

Research Paper (First-Author)

June 2025

- Conference on Data and Applications Security and Privacy (DBSec), Norway

SoK: Static Configuration Analysis in Infrastructure as Code Scripts

Published

Research Paper (First-Author)

August 2023

- IEEE International Conference on Cyber Security and Resilience (CSR), Venice, Italy

Access mechanism using inter planetary file system

Published

Research Paper (First-Author)

April 2019

- International Journal of Engineering and Technology, Volume 7, No 4, ISSN: 2227-524X

Securing Data in Cloud- A Physical Cyber System

Published

Research Paper (First-Author)

December 2017

- Journal of Computer Science Engineering, Volume-3, Issue-12, ISSN-2456-1843

Projects

AI for Human Centric Security | MBIE Catalyst Project

PhD Project

The University of Waikato

April 2022 - January 2026

- Developed a policy-driven code quality framework for Infrastructure as Code that detects simple vulnerabilities and complex threats such as supply chain attacks, performs advanced metadata-driven analysis to identify hidden security smell diffusion pathways, and evaluates AI-generated code quality and security using zero-shot approaches achieving 100% CIS security benchmark compliance versus 23.8% for human-written code.

Cryptographically Secure On-Line Identity System

Masters Project

Lancaster University

June 2020 - September 2020

- Funded by Quantum Base and The Royal Society, UK, this project developed a framework leveraging quantum computer-generated identities (PUFs) for device authentication and validation, targeting resource-constrained devices like IoT and mobile applications.

Secure communication using IPFS with IoT

Under Graduate Project

Amrita Vishwa Vidyapeetham

June 2018 - April 2019

- Funded by Amrita Vishwa Vidyapeetham, this project utilized the Interplanetary File System (IPFS) web3 protocol for secure, direct communication between users and IoT devices, eliminating the need for third-party vendors.

Crop Prediction System using Machine Learning algorithm

Industry Project

Amrita Vishwa Vidyapeetham

June 2017 - December 2017

- Funded by Amrita Vishwa Vidyapeetham, this project developed a K-Means-based Crop Prediction System to provide farmers with recommendations based on soil type, location, and sowing time.

Voluntary Activities

New Zealand Cyber Security Challenge

cybersecuritychallenge.org.nz

2022 - 2025

- Organized and facilitated Capture The Flag (CTF) events for the New Zealand Cyber Security Challenge, engaging over 500 participants, including high school students, university students, and industry professionals, to enhance their cybersecurity skills and awareness.

Cyber Security Workshop

Amrita School of Engineering, India

February 2016

- Organized and led a two-day workshop on Ethical Hacking and Cyber Security at Amrita School of Engineering, India, aimed at inspiring students from underprivileged backgrounds to explore opportunities in cybersecurity.

Interests

Hack The Box Active participant in Hack The Box challenges, honing penetration testing, problem-solving, and cybersecurity skills.

Travel

Passionate about traveling and exploring diverse cultures & learning about history, which has broadened my perspective and enriched my understanding of the world.

Video Games

Enthusiastic about gaming, particularly exploring immersive stories and unique experiences that games offer.

Swimming

Avid professional swimmer since the age of 10, fascinated by underwater exploration and challenges requiring breath control.